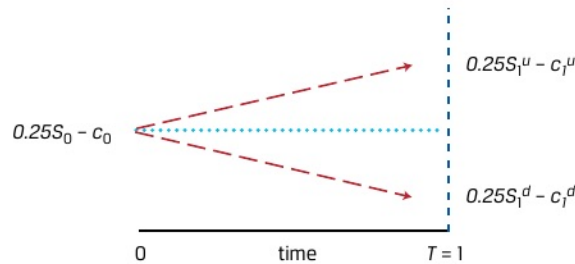


As can be seen, the value of the replicating portfolio is equal to CNY8 regardless of whether the price of the asset increases or decreases. Because of this, the portfolio is risk-free and can be discounted as a risk-free asset. The payoffs for these two scenarios are shown in Exhibit 16.

Exhibit 16: Call Option Replication



To solve for c_0 , we set the present value of the replicating portfolio ($0.25 \times 40 - c_0$) equal to the discounted future value of the risk-free payoff of CNY8 under each outcome as in Equation 5. Assume r of 5 percent:

$$V_0 = 0.25S_0 - c_0,$$

$$V_0 = \frac{V_1^u}{1+r} = \frac{V_1^d}{1+r}, \text{ and}$$

$$0.25 \times 40 - c_0 = \frac{\text{CNY}8}{1.05}.$$

We solve for c_0 as CNY2.38. Thus, CNY2.38 is a fair price for the seller of the contract to receive from the buyer of the contract.

As part of the example above, we assumed that the investor buys 0.25 units of the asset as part of the portfolio. This proportion of the underlying asset is known as the **hedge ratio** in option pricing. In fact, the example just completed is a simplified process for solving for the value of a call option (this material will be covered in more detail later in the curriculum). As in our prior examples, we compare different strategies using cash flow additivity to help solve for a no-arbitrage price for a financial instrument.

EXAMPLE 15

Put Option

A put option grants the owner the right, but not the obligation to sell a stock at a predetermined exercise price X . If we assume $X = \text{CNY}50$, then the put option value in one period (p_1) is equal to $\text{Max}(0, \text{CNY}50 - S_1)$.

1. Using a one-period binomial tree model with the same prices in one year, initial stock price (S_0) of CNY40, and 5 percent discount rate r , create a risk-free portfolio replicating the put option with 0.75 units of the underlying asset and solve for the put option price (p_0).

Solution:

The one-period binomial model is as follows: